# Smart Data Platform

Data Lakehouse - Executive Summary

Version 1.0
June 2026
Client Presentation

The Smart Data Platform is a modern data lakehouse that unifies operational and analytical workloads. It captures database changes in near real time, stores them as governed Iceberg tables on object storage, and delivers insights through standard SQL and self-service dashboards - built entirely on open-source, enterprise-grade technology.

## BUSINESS VALUE

**Near-Real-Time Analytics**
Operational data queryable within minutes, not overnight batches.

**Single Source of Truth**
Versioned Iceberg tables for consistent, auditable reporting.

**Lower Integration Cost**
Standard SQL and BI tools reduce specialized skill dependency.

**Future-Ready Foundation**
New domains onboard without architectural redesign.

## SOLUTION OVERVIEW

The platform replaces traditional batch ETL with Change Data Capture (CDC). Every insert, update, and delete in PostgreSQL is streamed through Kafka, materialized as Apache Iceberg tables, and made available for analytics - with full schema evolution, time travel, and ACID guarantees.

## ARCHITECTURE STACK

| Layer | Technology | Role |
|---|---|---|
| Source | PostgreSQL 16 | Operational database and CDC origin |
| Streaming | Apache Kafka 4.1 | Durable event backbone |
| Ingestion | Debezium + Iceberg Connect | CDC capture and lakehouse writes |
| Storage | MinIO (S3-compatible) | Columnar data files (Parquet) |
| Catalog | Project Nessie | Versioned table metadata |
| Compute | Trino 480 | Distributed SQL engine |
| BI | Apache Superset | Self-service dashboards |

## MEDALLION DATA LAYERS

| Layer | Purpose | Examples |
|---|---|---|
| Bronze | Raw CDC landing; full change history | merchants, transactions |
| Silver | Cleansed, deduplicated business entities | payments, merchant_risk |
| Gold | KPIs and executive reporting | daily_gmv, success_rate |

## PRODUCTION EVOLUTION PATH

| Component | Current (POC) | Production Target |
|---|---|---|
| Kafka | Single node | Multi-broker cluster (RF >= 3) |
| Object Storage | MinIO | AWS S3 / Azure / GCS |

| | | |
|---|---|---|
| Catalog | Nessie (in-memory) | Nessie with persistent backend |
| Query Engine | Single Trino node | Trino cluster with workers |
| Security | Development defaults | TLS, SSO, IAM, secrets vault |
| Orchestration | Docker Compose | Kubernetes with auto-scaling |

## KEY DIFFERENTIATORS

Open & Portable: 100% open-source; no vendor lock-in; production-ready on any cloud. Governed & Auditable: Iceberg snapshots, Nessie version branches, immutable Kafka event log. Scalable by Design: independent storage/compute scaling; containerized; clear POC-to-production path.